SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Total Marks: 25

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Code of Conduct

I R. Leela Krishna 192372363 (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Leela Krishna

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

(1) Define cloud Computing and explain and four characteristics that make it different from traditional on Premise Computing.

Cloud Computing is the delivery of computing resources such as servers, storage, database, software and networking over the Internet.

Four characteristics:-

(i) on-demand self-service:-

users can create or use computing resources whenever needed without direct assistance from the service provider.

(ii) Broad network access:-

Cloud services can be accessed through the Internet using devices such as laptops, smart phones and tablets.

(iii) Resource pooling:-

cloud provides share computing resources among multiple users using virtualization while maintain proper security and isolation.

(iv) Rabid elasticity :-

Resources can be quickly increase or decreased according to demand for example an online shopping website can add servers during a festival sales.

(2) Trace the evolution of cloud computing from hardware evolution to Internet software evolution and show how these changes enabled modern cloud platforms.

Cloud Computing developed through several important stages :-

Mainframe Computing -> Client Server -> Computing -> virtualization -> Internet Web servies -> Cloud Computing.

(i) mainframe Computing :-

In the early stage, many user shared a powerful central mainframe computer through terminals.

(ii) client - server Computing :-

Computing responsibilites were divided b/w Clients and servers. This allowed organized to provide

(iii) Virtualization :-

Virtualization allowed one physical server to run multiple virtual machines. This improved hardware utilization and reduced infrastructure cost.

(iv) :- modern cloud platforms :-

Cloud provider combined virtualization high-speed networking, distributed data centers, automation and web-services to provide scalable services such as IaaS, PaaS and SaaS

(3) Explain the role of server virtualization in cloud scalable services such as IaaS, PaaS, and SaaS. as an enabling technology and cloud computing as a service model.

Role of server virtualization

Server virtualization divides one physical server into multiple virtual machines (VMs) using a software layer called a hypervisor.

for example

Physical server

↓

Hypervisor → VM1 + VM2 + VM3

Benefits :-

* Improve in physical server utilization
* Reduces hardware and electricity costs
* makes resources allocation flexible.
* Supports easy scaling and vim migration.

virtualization

* Technology used to create virtual resources
* uses hypervision and virtual machines
* ex :- vmware virtual machine.

Cloud computing

* service model for delivering computing resources.
* uses data center networks virtualization and automation
* ex :- Amazon EC2

(4) Discuss the role of data centers in delivering cloud services.

A data centre is a facility containing servers storage system, networking equipment, power used to provide computing services they are five types.

Roles of data centers.

(i) :- Computing resources :-

Data centers contain powerfull servers that run cloud applications and virtual machines

ii) Data storage :-

They store user-file, databases, backups and application data.

(iii) Networking

High-speed networks connect servers and provide Internet access to users.

(iv) availability and reliability :-

Backup power, redundant hardware, and multiple network connections help keep services available

(v) Scalability :-

cloud providers can add more servers and storage when user demand increases.

(5) Differentiable laas paas, saas and xaas withone suitable real world example for each model.

model	meaning	wait it	example.
Iaas	Infrastructure as a service	provides virtual machines	Amazon EC2.
Paas	Platform as a service	Development platforms	google APP Engine.
Saas	Software as a service	Read to use Software through Internet	google DOCS
Xaas	Anything as a service	different IT resources delivered as a service	Amazon RDS

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks: